

Networking Basics for AI Infrastructure: VPCs, Firewalls, Load Balancers

A comprehensive guide to building secure, scalable networking foundations for your AI/ML workloads



Why Networking Matters in AI Infrastructure

Modern AI systems are inherently **distributed** across multiple nodes and services, requiring robust networking to function effectively.

Data Transfer & Communication

Enables massive training datasets to move efficiently between storage and compute resources

Security Foundation

Controls who can access your sensitive models, datasets, and inference endpoints

Performance Critical

Low latency connections dramatically impact model training speed and inference responsiveness



Virtual Private Cloud (VPC): Your AI Network Foundation

A VPC creates an **isolated section of the cloud provider's network** that you control, serving as the foundation for your AI infrastructure.

1

IP Address Management

Define custom CIDR blocks (IP ranges) to organize your ML infrastructure

2

Network Topology Control

Create subnets, route tables, and gateways to structure traffic flow

3

Security Boundaries

Isolate sensitive AI workloads from other systems and the public internet

4

Cloud-Provider Integration

Connect to managed AI services while maintaining security controls

Subnets: Logical Divisions Within Your VPC

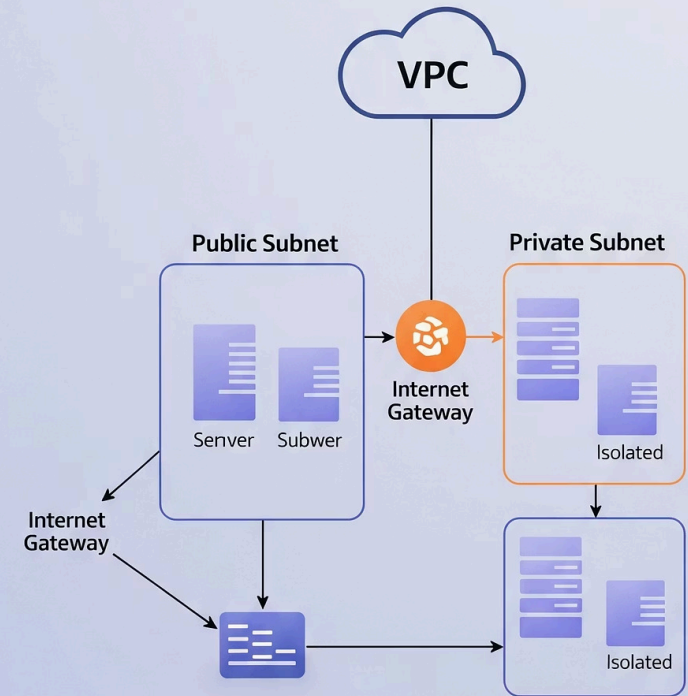
Public Subnets

- Connected to internet gateway
- Host public-facing ML APIs
- Run inference endpoints
- Host management components

Private Subnets

- No direct internet access
- Host training clusters
- Store sensitive datasets
- Run internal processing jobs

This separation creates a **defense-in-depth strategy** for your AI infrastructure, limiting exposure of your most sensitive components.



Firewalls: First Line of Defense for AI Workloads

Firewalls act as filters for all traffic entering and exiting your ML infrastructure, enforcing precise access controls.

Rule-Based Control

Specify exactly which traffic is allowed/denied based on IP addresses, ports, and protocols

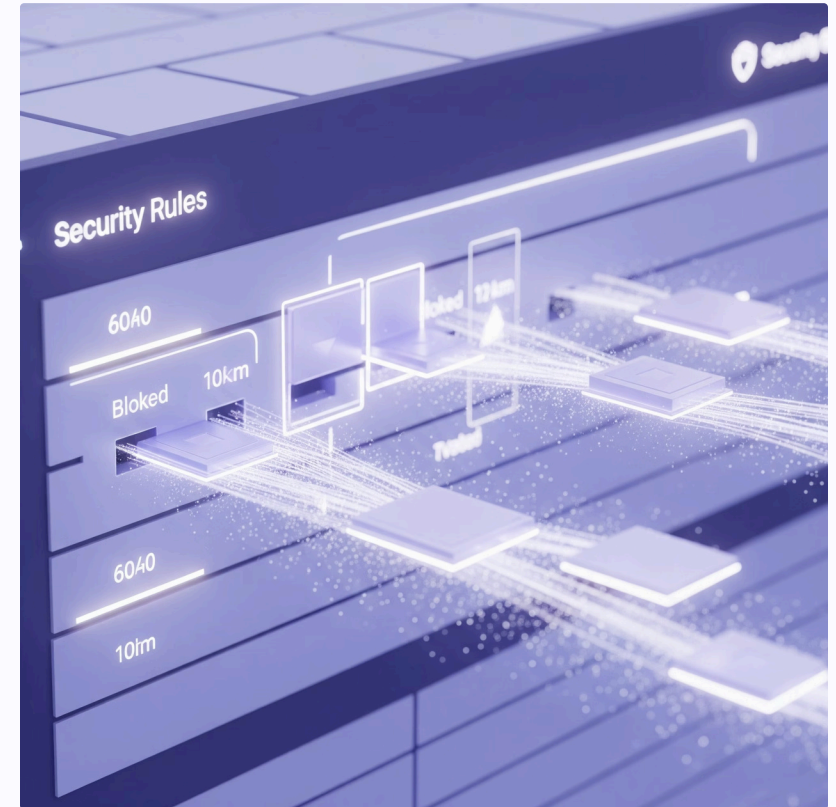
ML Infrastructure Protection

Prevent unauthorized access to training data, model weights, and inference endpoints

Multi-Tenant Isolation

Essential for shared GPU clusters serving multiple teams or customers

Example: Allow SSH (port 22) *only* from your team's IPs while blocking all other incoming traffic to training nodes



Security Groups vs. Network ACLs

Security Groups

- **Instance-level** firewall protection
- **Stateful** - return traffic automatically allowed
- Support allow rules only
- Applied to specific instances
- Evaluated collectively before traffic reaches instance

Network ACLs

- **Subnet-level** firewall protection
- **Stateless** - return traffic requires explicit rules
- Support both allow and deny rules
- Applied to all instances in subnet
- Evaluated in numbered order (first match wins)

Combining both provides **defense in depth** for your AI workloads, with NACLs as broad subnet protection and security groups for fine-grained control.



Load Balancers: Why They're Critical for AI Systems

Load balancers intelligently distribute requests across multiple servers, enabling scalable AI systems that can handle varying workloads.

Horizontal Scaling

Allows you to add more inference servers as demand increases, rather than scaling vertically

High Availability

Automatically routes around failed nodes, maintaining ML service uptime

Traffic Management

Intelligently routes requests based on server health, capacity, and response times

Inference Endpoint Scaling

Enables auto-scaling groups to handle variable prediction workloads efficiently

Types of Load Balancers for AI Workloads

By Protocol Layer

Layer 4 (Transport)

- Routes based on IP address and port
- Handles TCP/UDP traffic
- Lower overhead, higher throughput
- Great for raw traffic distribution

Layer 7 (Application)

- Routes based on HTTP/HTTPS headers
- Can direct traffic by URL path
- Supports content-based routing
- Ideal for complex API endpoints

By Network Position

Internal Load Balancers

- Distribute within private subnets
- Not accessible from internet
- For microservices communication
- Used between processing stages

External Load Balancers

- Public-facing endpoints
- Distribute internet traffic
- Protect from DDoS attacks
- Front public AI APIs

Networking for AI Workloads: Special Considerations



Training Workloads

Require **ultra-fast internal networks** like InfiniBand, RDMA, or NVLink for efficient distributed training

- Typically run in private subnets
- Need high bandwidth, low latency connections
- Often use specialized network topologies

Inference Workloads

Need **public endpoints with load balancing** to serve predictions at scale

- Exposed via public subnets
- Protected by WAFs and API gateways
- Require flexible scaling capabilities

Networking Best Practices for AI Infrastructure

1

Restrict Administrative Access

Limit SSH access to your infrastructure to specific IPs only, using bastion hosts for added security

2

Protect Databases and Storage

Never expose your ML databases or data storage directly to the internet; access through private subnets only

3

Encrypt All Traffic

Use HTTPS with proper certificates for all inference APIs, and encrypt data in transit between components

4

Monitor Network Traffic

Implement VPC Flow Logs and traffic analysis to identify unusual patterns that might indicate security issues

5

Infrastructure as Code

Automate network configurations with Terraform or CloudFormation to ensure consistency and reproducibility

Key Takeaways: Networking for AI Infrastructure



VPCs are your foundation

Create isolated, customizable cloud networks that serve as the secure foundation for all AI workloads



Defense in depth

Combine firewalls at subnet and instance levels to create comprehensive security for sensitive models and data



Scale with load balancers

Implement load balancers to enable reliable, scalable inference endpoints that can handle variable demand

Strong networking architecture directly impacts the **performance, security, and cost-efficiency** of your AI infrastructure

